

|     |     |                                                                                                                                                                                                                                                                                                                                            |                |     |
|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|
| 7   | (a) | <p><i>either</i> <math>V = E R_1 / (R_1 + R_2)</math>    <i>or</i>    <math>I = E / (R_1 + R_2)</math></p> <p><math>= \frac{1800}{3000} \times 4.50</math>                      <math>V = \frac{1800}{3000} \times 4.50</math></p> <p><math>= 2.70 \text{ V}</math>                                      <math>= 2.70 \text{ V}</math></p> | C1<br>M1<br>A0 | [2] |
| (b) | (i) | <p>for a wire, <math>V = I \times (\rho L/A)</math></p> <p><math>I</math>, <math>\rho</math> and <math>A</math> are constant</p> <p>so <math>V \propto L</math></p>                                                                                                                                                                        | M1<br>A1<br>A0 | [2] |

| Page 3 | Mark Scheme                    | Syllabus | Paper |
|--------|--------------------------------|----------|-------|
|        | GCE A/AS LEVEL – November 2005 | 9702     | 2     |

(ii) 1 2.70 V A1 [1]

2  $\frac{L}{100} = \frac{2.70}{4.50}$  C1

$L = 60.0$  cm A1 [2]

(iii) thermistor resistance decreases as temperature rises M1  
so QM is shorter A1 [2]

8 (a) product of force and distance

M1